Tanay Ande | AI/ML Engineer

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Summary

AI/ML Engineer with 4 years of experience designing, developing, and deploying scalable Machine Learning, Deep Learning, NLP, and Generative AI solutions across financial analytics, enterprise AI, and cloud platforms. Experienced in building end-to-end ML pipelines, predictive analytics systems, real-time data processing frameworks, and LLM-powered applications using Python, PySpark, Azure Databricks, TensorFlow, Scikit-learn, Hugging Face, and Kubernetes. Strong expertise in MLOps, model deployment, feature engineering, distributed computing, and cloud-native AI architectures on Azure and AWS. Proven ability to improve model accuracy, automate workflows, optimize business operations, and deliver production-grade AI solutions in enterprise environments.

Experience

Scale AI, AI/ML Engineer **USA | 02/2024 – Present**

- Developed enterprise-grade AI/ML solutions leveraging Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Transformer-based architectures to automate intelligent workflows and enhance AI-driven decision-making systems.
- Built scalable distributed ML pipelines using Python, PySpark, Spark Structured Streaming, and Databricks, processing high-volume structured and unstructured datasets with improved operational efficiency and reduced inference latency.
- Designed and fine-tuned NLP models using Hugging Face Transformers, Sentence-BERT, and RoBERTa for semantic search, document classification, entity extraction, and contextual intelligence applications.
- Implemented end-to-end MLOps workflows using MLflow, Kubeflow, Docker, Kubernetes, and KServe for automated training, deployment, monitoring, and versioning of machine learning models in production environments.
- Integrated vector databases and embedding pipelines for semantic retrieval systems supporting GenAI and enterprise search applications with high accuracy and scalability.
- Optimized model performance using Optuna hyperparameter tuning, SHAP explainability, and feature engineering techniques, improving prediction accuracy and model interpretability.
- Collaborated with cross-functional engineering and product teams to deploy AI services on Azure and AWS cloud infrastructure, ensuring scalability, reliability, and high availability of ML systems.
- Built real-time monitoring and drift detection frameworks using Evidently AI and custom analytics dashboards, maintaining production model accuracy and operational stability.

S&P Global, AI Engineer/Data Scientist **India | 09/2020 – 04/2022**

- Developed an Intelligent Financial Analytics Platform to process and analyze large-scale market and transaction datasets, improving operational efficiency by 22% and enabling faster data-driven business decisions for investment and risk analysis teams.
- Built scalable ETL and streaming data pipelines using Azure Databricks, Delta Lake, Spark Structured Streaming, and PySpark to integrate structured financial datasets with unstructured market intelligence for advanced analytics applications.
- Designed and deployed machine learning and statistical models using LightGBM, Transformer-based Autoencoders, and Scikit-learn for forecasting, anomaly detection, and financial instrument classification, achieving 89% F1-score and 94% recall.
- Applied NLP and deep learning techniques using RoBERTa, Sentence-BERT, and Logistic Regression to extract actionable insights from financial reports, analyst notes, and transaction logs, reducing manual analysis efforts by 21%.
- Performed feature engineering, Bayesian hyperparameter optimization using Optuna, and model explainability using SHAP and LIME to improve prediction accuracy, transparency, and stakeholder confidence in AI-driven insights.
- Containerized machine learning applications using Docker and deployed scalable inference services on Azure Kubernetes Service (AKS) with KServe, enabling real-time analytics and integration with enterprise dashboards and reporting systems.
- Implemented automated MLOps workflows using MLflow, Kubeflow, and Evidently AI for continuous training, monitoring, drift detection, and model lifecycle management, maintaining production model accuracy above 91%.

Skills

- **Programming & Query Languages:** Python, SQL, PySpark, Scala, Shell Scripting
- **Machine Learning & AI:** Machine Learning, Deep Learning, NLP, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Forecasting, Anomaly Detection, Recommendation Systems, Classification, Feature Engineering, Model Optimization, OpenAI
- **AI/ML Frameworks & Libraries:** Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, Hugging Face Transformers, Sentence-BERT, RoBERTa, Optuna, SHAP, LIME

- **Data Engineering & Big Data:** Apache Spark, Spark Structured Streaming, Delta Lake, Azure Databricks, Kafka, ETL Pipelines, Real-Time Data Processing
- **MLOps & Deployment:** MLflow, Kubeflow, Docker, Kubernetes, KServe, CI/CD Pipelines, Model Monitoring, Evidently AI,FastAPI
- **Cloud & Platforms**: Microsoft Azure, Azure Kubernetes Service (AKS), Azure Data Factory, AWS Cloud Services
- **Databases & Analytics:** Vector Databases, PostgreSQL, MySQL, MongoDB
- **Tools & Collaboration:** Git, GitHub, Jira, Agile/Scrum Methodologies

Projects

Enterprise RAG Q&A System | LangChain, Qdrant, FastEmbed, Flashrank, OpenAI GPT-4, FastAPI, Python

- Built a production-grade document QA system parsing and semantically indexing financial reports using LangChain + Qdrant with hybrid search (dense + sparse) and re-ranking via Flashrank, improving retrieval precision by 30% over baseline.
- Achieved sub-300ms end-to-end response latency; exposed via FastAPI endpoints with async processing and Redis caching layer for high-traffic use cases.
- Translated complex model outputs and retrieval results into clear, actionable dashboards for operations and executive stakeholders, enabling data-driven decisions across business units.

Agentic AI Customer Support Pipeline | *LangGraph, Groq AI, FastAPI, SQLite, Python*

- Designed a multi-step agentic RAG pipeline using LangGraph for workflow orchestration and Groq AI for ultra-low latency LLM inference (~100ms), handling 1K+ queries/day with 95%+ resolution rate.
- Implemented conversation memory, tool-use agents, and fallback escalation logic; integrated into enterprise workflows via FastAPI REST endpoints.
- Designed pipeline to support real-time data ingestion from MQTT-based event streams, enabling generative AI integration into enterprise operational environments with sub-second responsiveness.

EDUCATION

Master |University of Maryland MIS (management information systems) **May 2022 - May 2024**
Bachelor |Loyola Academy Hyderabad **Aug 2018 – Jul 2021**